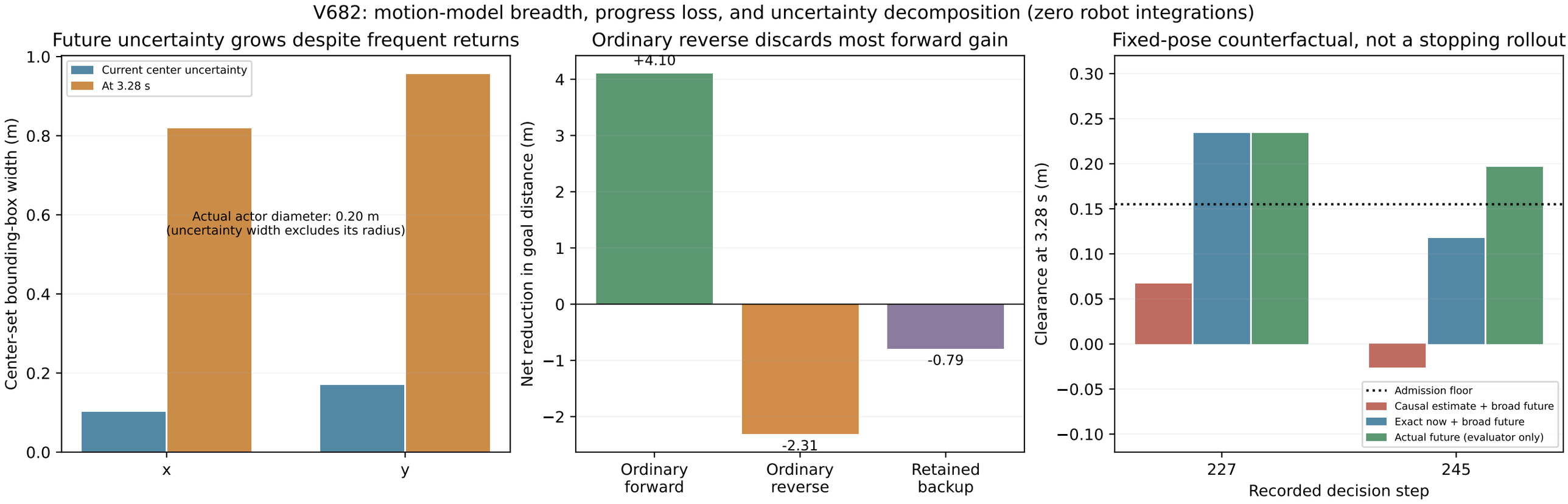
V682: motion-model breadth, progress loss, and uncertainty decomposition (zero robot integrations)
Future uncertainty grows despite frequent returns
Current center uncertainty
At 3.28 s
Actual actor diameter: 0.20 m
(uncertainty width excludes its radius)
Center-set bounding-box width (m)
1.0
0.8
0.6
0.4
0.2
0.0
x
y
Ordinary reverse discards most forward gain
+4.10
-2.31
-0.79
Net reduction in goal distance (m)
4
3
2
1
0
−1
−2
Ordinary forward
Ordinary reverse
Retained backup
Fixed-pose counterfactual, not a stopping rollout
Clearance at 3.28 s (m)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
Admission floor
Causal estimate + broad future
Exact now + broad future
Actual future (evaluator only)
227
245
Recorded decision step